A Memory System for the Storage and Retrieval of Parametric and Symbolic Data Representations

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Title: A Memory System for the Storage and Retrieval of Parametric and Symbolic Data Representations

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Cross-Reference to Related Applications

This application is related to and builds upon the concepts disclosed in the U.S. Patent Applications for "A Central Processing Unit Architecture for Native Parametric and Symbolic Computation" and "Method and System for High-Radix Data Representation Using Parametric Mathematical Encoding." The present invention provides a memory architecture necessary for the persistent storage of the data formats utilized by said inventions, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A memory system and architecture designed for the storage and retrieval of data as parametric or symbolic representations rather than as fixed-width, evaluated numerical values. The system comprises a memory controller and a storage medium (such as DRAM, flash, or ROM) where data is stored as variable-length structures representing mathematical 'recipes'

or expressions. The memory controller manages the dynamic allocation of physical memory to accommodate these structures and interfaces with a parametric data bus. This architecture ensures that the precision of data is not lost when it is moved from a parametric CPU to memory, enabling the persistence of perfectly represented fractional, irrational, and large-magnitude numbers throughout a complete computing system.

Background of the Invention

Conventional computer memory systems, including RAM and non-volatile storage, are architected to store data in fixed-size blocks or words (e.g., 8-bit bytes, 64-bit words). Data is accessed via a memory address that points to a specific physical location containing a discrete, binary-encoded value. This model forces any number to be approximated or truncated to fit within the predefined word size of the memory architecture. When a parametric CPU, which can handle numbers with perfect precision, needs to store data, it must first convert its symbolic representation into a fixed-width, lossy format. This act of "saving to memory" becomes the primary bottleneck where the benefits of parametric computation are lost.

Summary of the Invention

The present invention provides a memory system that natively stores numbers and constants in their parametric, symbolic form, thereby creating a complete, end-to-end parametric computing environment. The system fundamentally changes the paradigm of data storage from saving discrete values at addresses to storing unevaluated mathematical expressions in a managed space.

The system is comprised of a Parametric Memory Controller and a storage medium. The Parametric Memory Controller is the core innovation. It receives symbolic data structures (e.g., "recipes" for numbers like '7/16' or 'Pi') from a parametric CPU or data bus. Instead of storing a single evaluated value, the controller allocates a variable amount of physical memory space required to hold the entire data structure for the expression. It maintains an index or addressing scheme to locate and retrieve these variable-length structures.

When a read request is received, the controller retrieves the complete, unevaluated data structure from the storage medium and transmits it back to the CPU. This ensures that data remains in its perfect, symbolic form whether it is in the CPU's registers, on the data bus, or resting in memory. This architecture can be applied to any memory type, including volatile

RAM for active computations, non-volatile memory for persistent storage of high-precision datasets, or Read-Only Memory (ROM) for factory-provisioning of perfectly represented mathematical and physical constants.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the architecture of the parametric memory system and its interaction with a parametric CPU.

FIG. 2 is a conceptual diagram illustrating how a parametric expression is stored as a variable-length data structure within the physical memory medium.

Detailed Description of the Invention

Referring to **FIG. 1**, the parametric memory system (100) is shown in context with a parametric CPU (150). A Parametric Memory Controller (110) serves as the intermediary between the CPU and the physical memory storage (120). When the CPU (150) issues a 'write' command, it places a symbolic data structure (160) onto the data bus. The Parametric Memory Controller (110) receives this structure. It determines the size of the structure and allocates a sufficient block of physical memory (130) within the storage medium (120) to hold it. The controller then writes the structure to this block and updates its internal address index. For a 'read' command, the controller uses its index to locate, retrieve, and transmit the complete, unmodified symbolic structure back to the CPU.

FIG. 2 provides a conceptual view of the storage medium (200). It contrasts the traditional method of storing an evaluated number (210), where the fraction '7/16' is converted to

a lossy, fixed-width floating-point representation, with the parametric method (220). In the parametric method, the memory controller allocates a variable-length block (230) that is precisely large enough to hold the entire symbolic structure, with distinct fields for 'Operand: 7', 'Operator: /', and 'Operand: 16'. This preserves the number's absolute precision within the memory system itself.

Claims

What is claimed is:

1. A memory system for storing data, comprising:
    - a physical storage medium; and
    - a parametric memory controller configured to:
        - receive a data structure representing a mathematical expression of a number;
        - dynamically allocate a variable-length block of the physical storage medium to store said data structure; and
        - retrieve the complete, unmodified data structure from the storage medium upon a read request.
2. The memory system of claim 1, wherein the data structure is received from a parametric central processing unit over a parametric data bus.
3. The memory system of claim 1, wherein the mathematical expression represents a fractional number, and said representation is stored without loss of precision.
4. The memory system of claim 1, wherein the mathematical expression is a symbolic representation of a mathematical or physical constant.
5. The memory system of claim 1, wherein the parametric memory controller manages an addressing index to locate the variable-length blocks within the physical storage medium.
6. The memory system of claim 1, wherein the physical storage medium is volatile random-access memory (RAM).
7. The memory system of claim 1, wherein the physical storage medium is non-volatile memory.

8. The memory system of claim 7, wherein the physical storage medium is read-only memory (ROM) pre-configured with a library of data structures representing mathematical constants.

9. A method for storing data, comprising:

    - receiving, at a memory controller, a data structure representing a mathematical expression;
    - determining the size of the received data structure;
    - allocating a block of memory in a storage medium, wherein the size of the block is based on the determined size of the data structure;
    - storing the entire data structure in the allocated block; and
    - creating an index entry that maps a memory address to the location of the allocated block.

Drawings

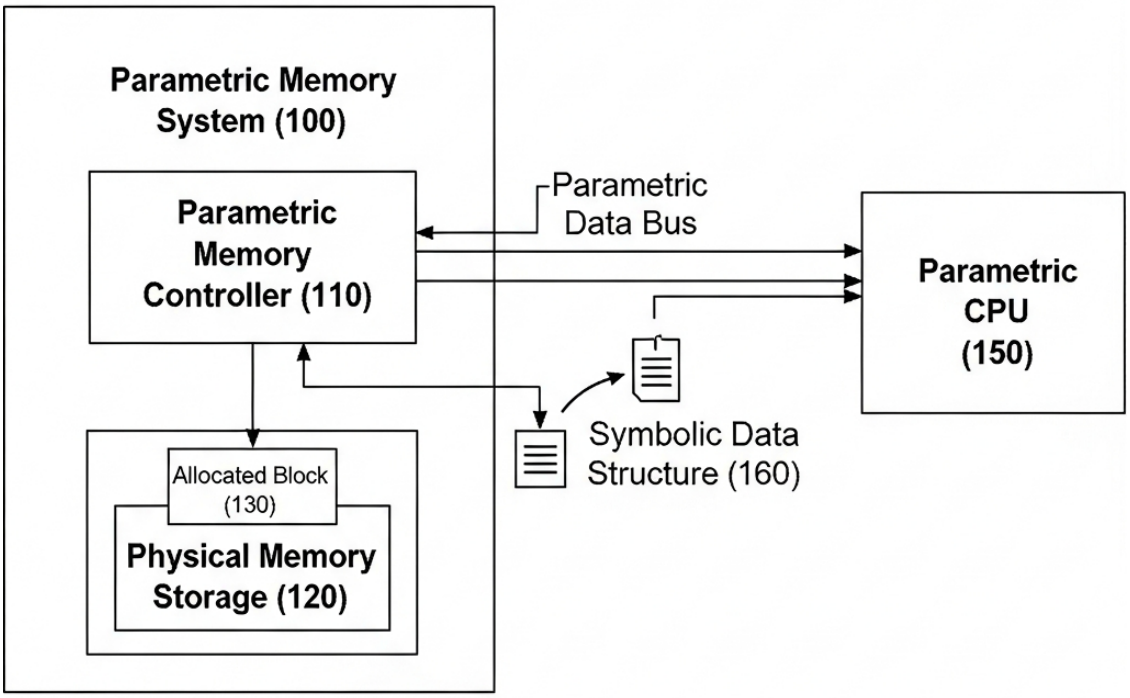


Figure 1: Block diagram illustrating the architecture of the parametric memory system and its interaction with a parametric CPU.

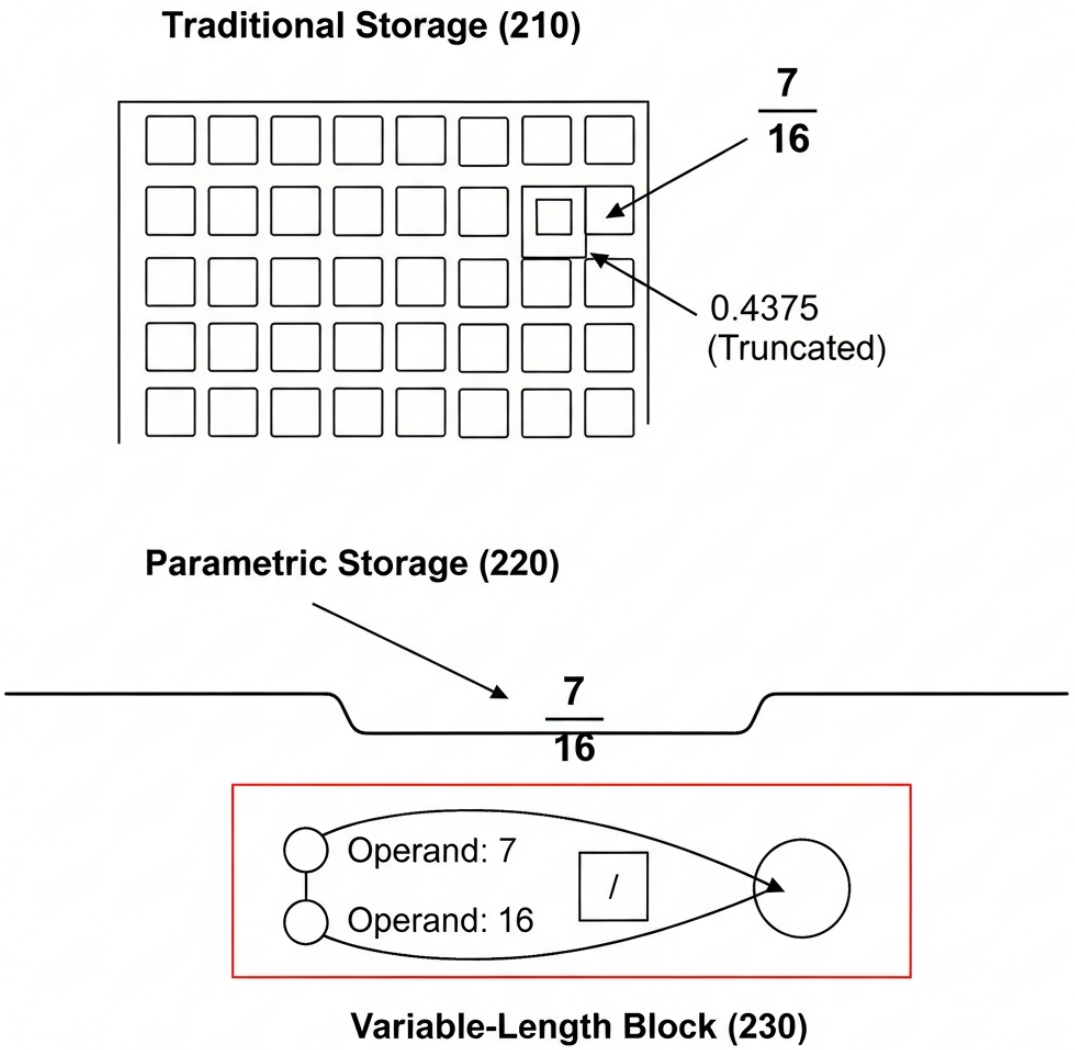


Figure 2: Conceptual diagram illustrating how a parametric expression is stored as a variable-length data structure within the physical memory medium.